

TOPIC 1/60

AI vs ML vs Deep Learning vs GenAI

Demystifying the buzzwords.

What's the actual difference?

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THE BIG PICTURE

Think of them as **Russian nesting dolls**. Each technology is a subset of the one before it.

AI (The Universe)

↳ ML (The Galaxy)

↳ Deep Learning (The Solar System)

↳ **GenAI (The Earth)**

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THE BROADEST SCOPE



Artificial Intelligence (AI)

Any technique that enables computers to mimic human behavior or intelligence.

Example: A chess bot from the 1990s. It doesn't "learn"; it just follows thousands of pre-programmed "if-then" rules written by human coders.

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THE NEXT LAYER

Machine Learning

(ML)

A subset of AI. Instead of explicitly programming the rules, you give the machine data and let it **learn** the rules itself.

"Show, don't tell."

Show a computer 10,000 photos of cats and 10,000 photos of dogs. It figures out the mathematical patterns to tell them apart.

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THE BRAIN-INSPIRED LAYER

Deep Learning

(DL)

A subset of ML that uses multi-layered **Artificial Neural Networks** inspired by the human brain.

Needs massive amounts of data and computing power (GPUs). It's the technology behind self-driving cars and voice assistants like Siri.

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THE LATEST BREAKTHROUGH

Generative AI

(GenAI)

A subset of Deep Learning. Instead of just *analyzing* or *predicting* data, it **creates entirely new data**.

It creates from scratch:



Text: ChatGPT, Claude



Images: Midjourney, DALL-E



Audio: Suno, Udio

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ANALYSIS VS CREATION

TRADITIONAL ML

"Is this a picture of a cat? Yes or No."

> It classifies existing data.

GENERATIVE AI

"Draw me a picture of a neon cat riding a skateboard on Mars."

> It generates novel data.

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PUTTING IT TOGETHER

The Stack

How these concepts relate in the modern tech ecosystem:

AI = The Goal

ML = The Method

DL = The Engine

GenAI = The Output

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WHY GENAI NOW?

Massive Computing

Modern GPUs (like Nvidia's) finally have the power to train these massive models.

Transformers

A breakthrough neural network architecture invented in 2017 that revolutionized natural language.

Ocean of Data

The public internet provided trillions of words and images to use as training material.

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